

Predicting Arbitration Salaries in Major League Baseball

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Predictive Modeling in Sports

Overview



Machine learning model developed to predict Major League Baseball salary arbitration outcomes



Generated an estimate of what MLB arbitration market would award a player



Purely based on the statistical patterns observed from 2023-2025 training data



Finds best predictors of dollar value in the MLB arbitration market

MLB Financial Distinctions

Luxury Tax System, No Salary Cap

Requires teams spending above a certain threshold to pay an additional tax, does not place a limit on spending

Market Size Matters

Wide variation in team spending according to available resources

Club Success Depends on Financial Efficiency

Efficient payroll and accurate player valuations lead to long-term competitiveness

Financial Stages of MLB Career



Baseball Arbitration Hearing Process

1

Lack of Contract Agreement

Club and arbitration-eligible player cannot agree on a contract for the upcoming season

2

Salary Figures Filed

Each side submits one salary figure to an independent arbitration panel, one of which will be selected

3

Hearing

Both sides argue their case using statistics, player “comps”, and performance narratives

4

Panel Decision

Panel must choose one of the filed figures, no in between

Literature Review Takeaways

Machine Learning Methods

- Regression and clustering algorithms successfully identify undervalued and overvalued players
- Most prior ML work targets free agents, so a clear modeling gap exists for arbitration-specific salary prediction

Player Agents

- PCA resolves multicollinearity among highly correlated performance stats
- Pre-arbitration metrics are significant predictors of first-year arbitration salary
- Consistency across pre-arb years adds explanatory power beyond raw counting statistics alone

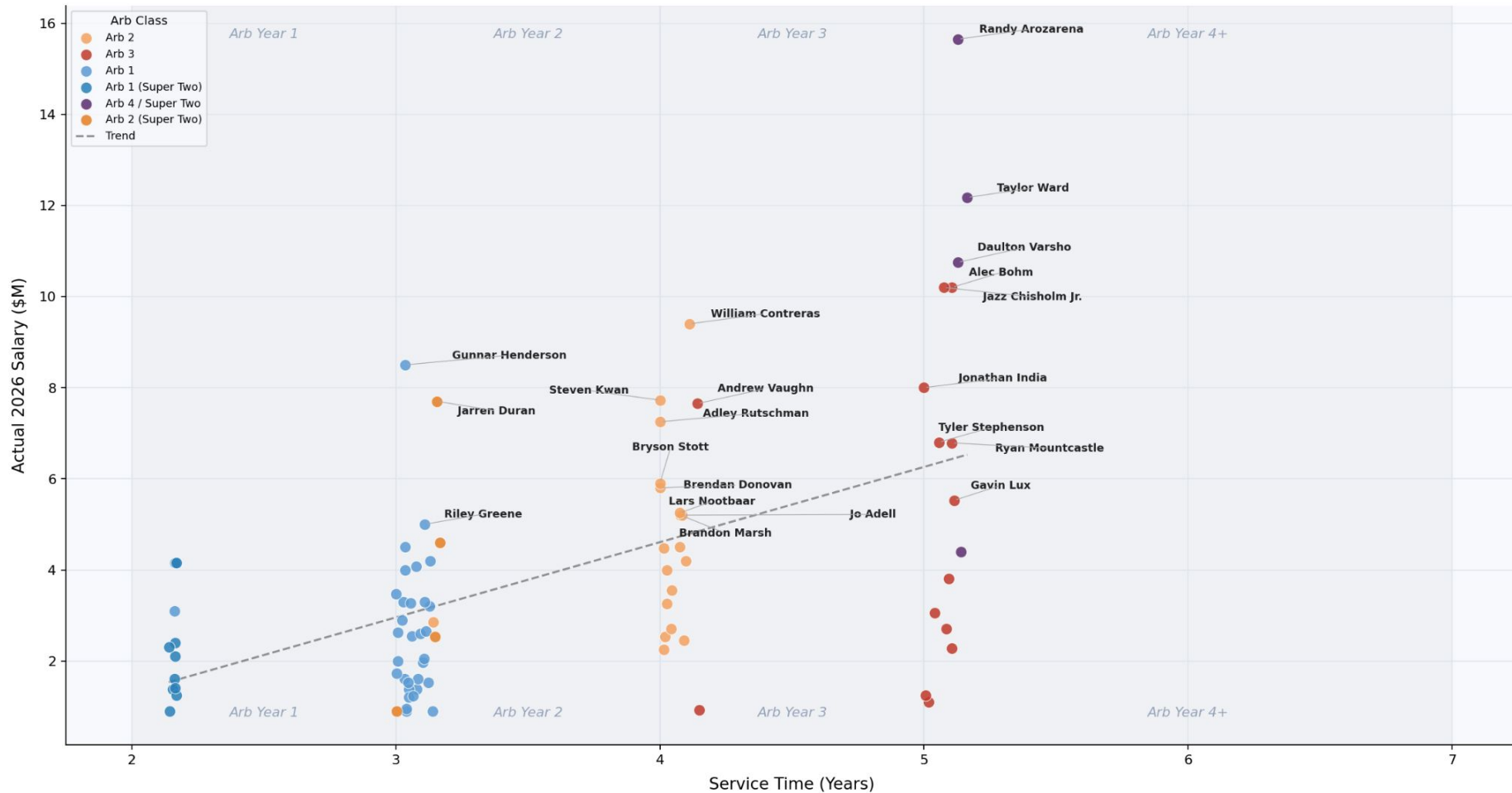
Arbitration Inequality

- Arbitrator awards are subjective, player and team filings are strategic guesswork
- Arbitrator inconsistency makes outcomes unpredictable
- OLS baseline models can quantify underpayment and expose systemic salary gaps

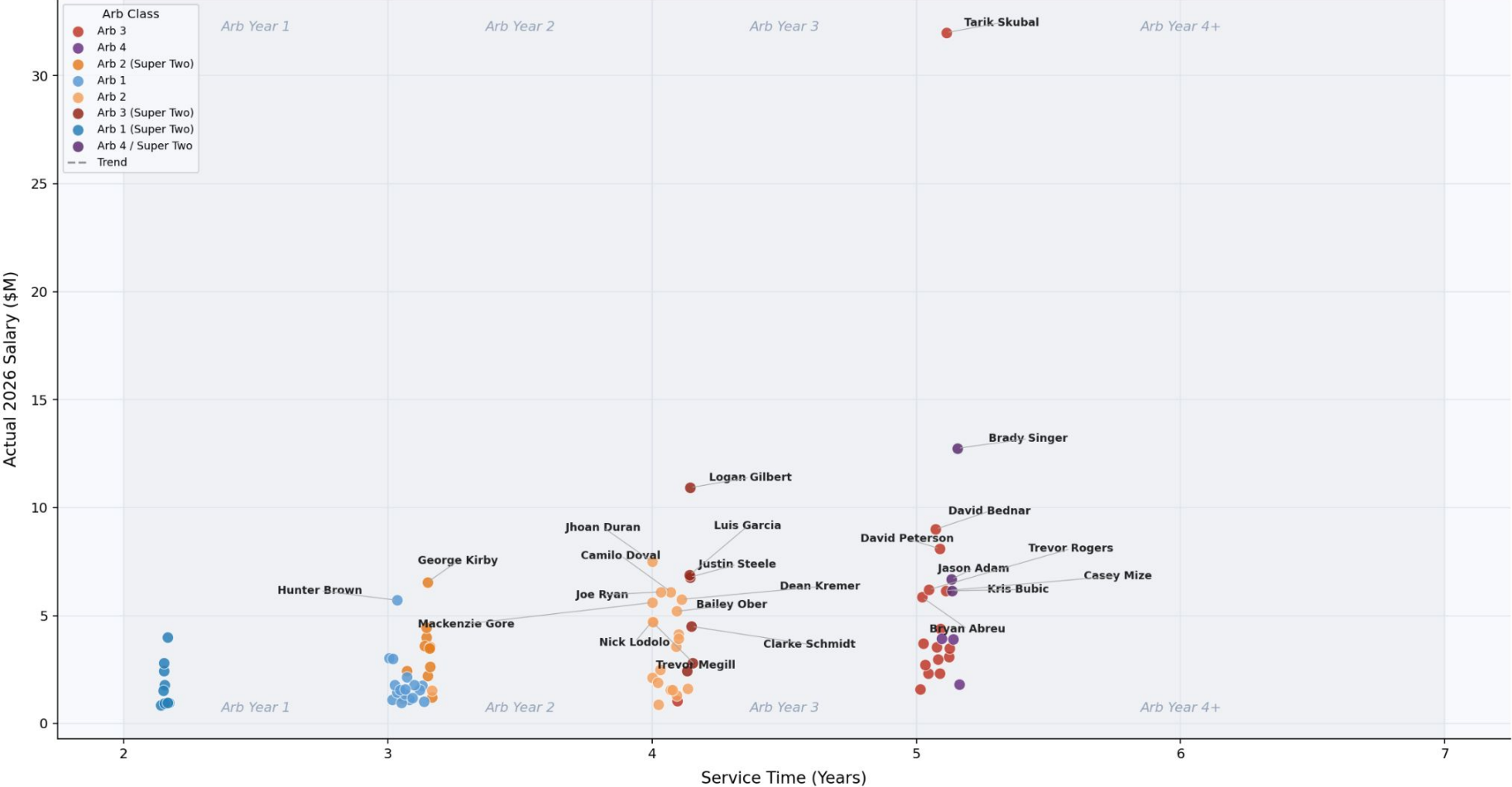
Distribution of Arbitration Salaries 2023-2025 Training Data (n=609)



Service Time vs. Salary — Hitters (2026 Arbitration)



Service Time vs. Salary — Pitchers (2026 Arbitration)



Data Sources

FanGraphs

Provides many traditional baseball metrics, sometimes referred to as “counting stats”

Baseball Savant

Much further context through publicly available Statcast metrics

Spotrac

Key information on contract status, service time, and salary history

Model Features

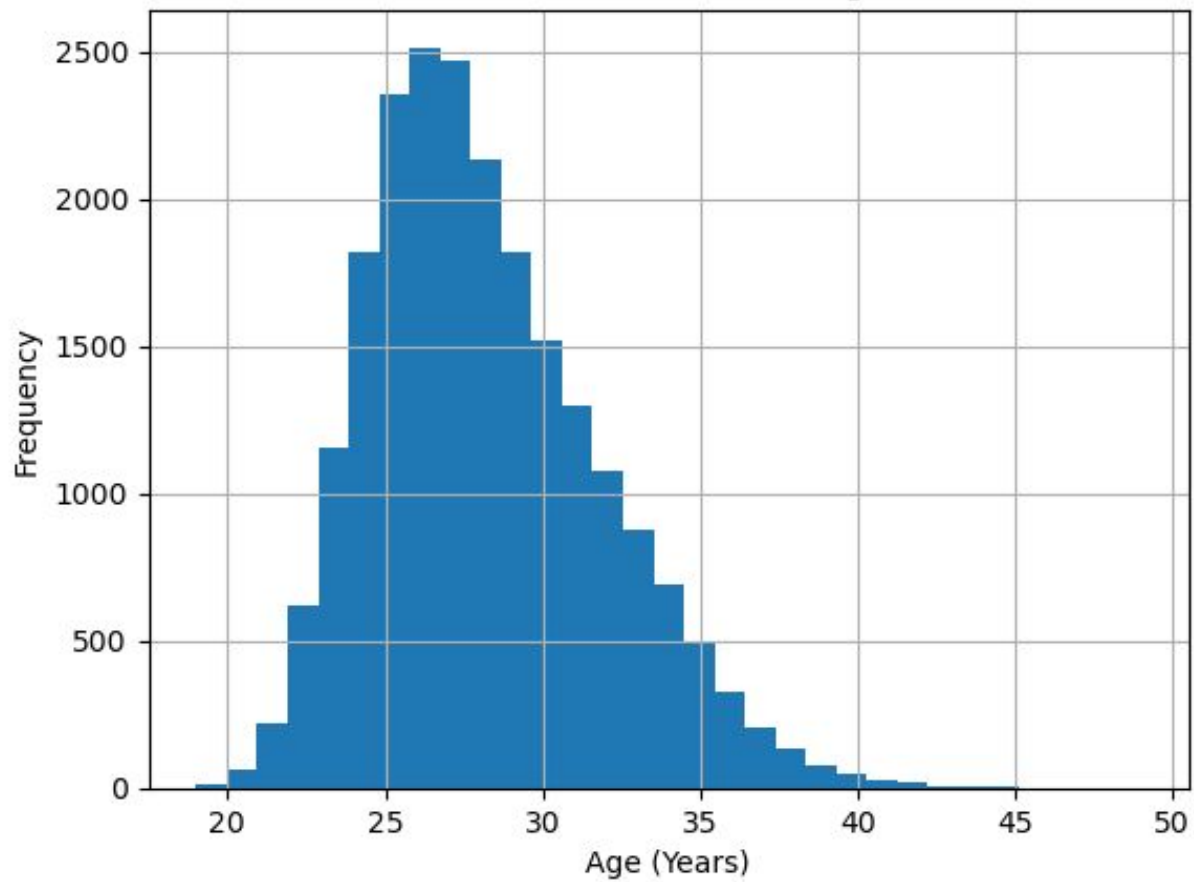
Hitters (56 Numeric, 2 Categorical)

- Platform-year AVG, HR, OPS, WAR, wRC+, etc
- 3-year rolling aggregates
- Career totals and averages
- Statcast: exit velo, barrel %, xwOBA, sprint speed
- Age, service time, arb class, position in field

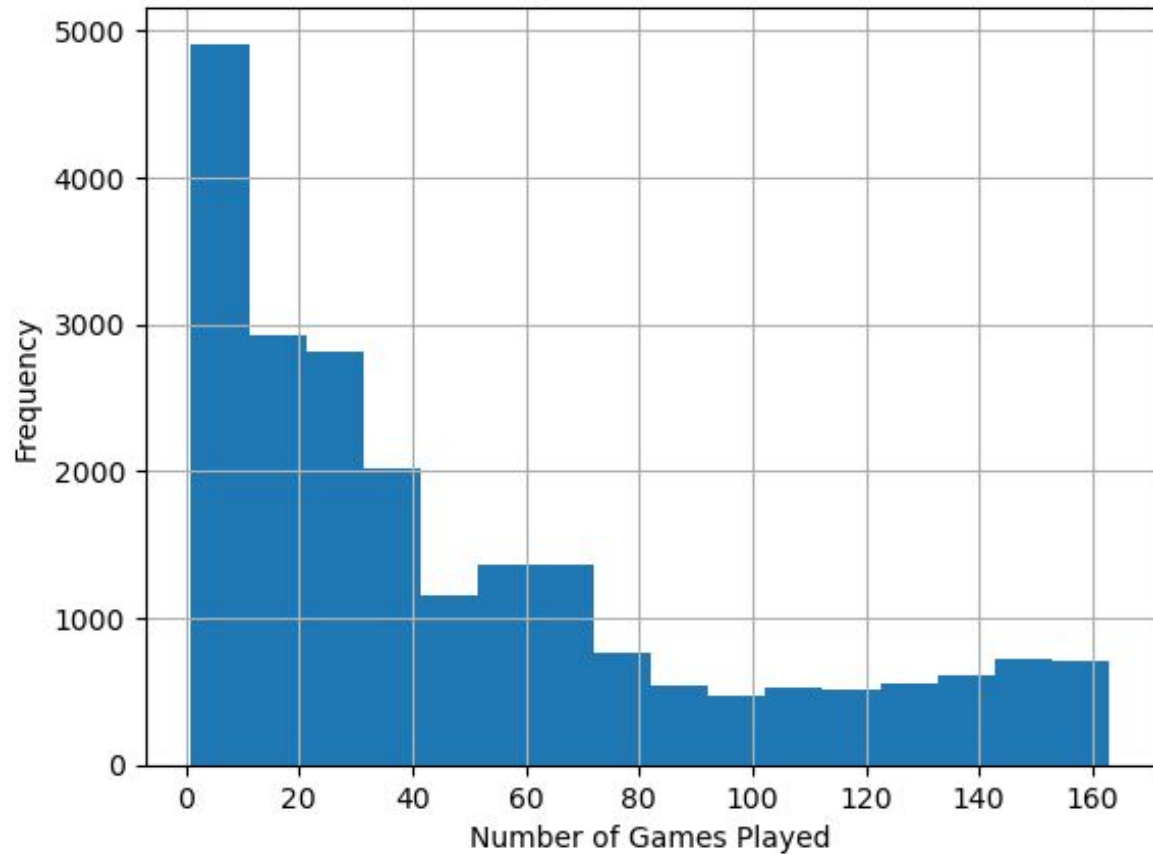
Pitchers (50 Numeric, 2 Categorical)

- Platform-year ERA, FIP, SIERA, K%, HR/9, etc
- 3-year rolling aggregates
- Career totals and averages
- Statcast: xERA, exit velo against, barrel % against
- Age, service time, arb class, role (SP/RP)

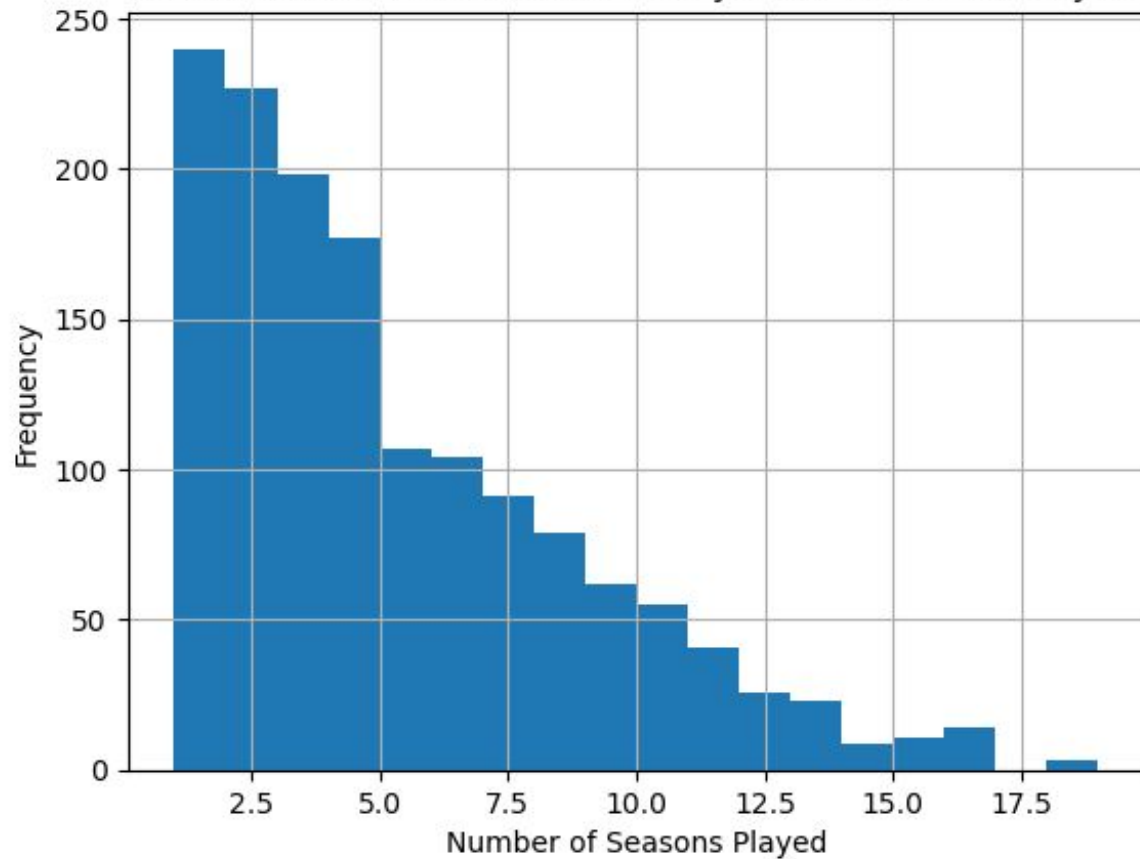
Distribution of Hitter Age



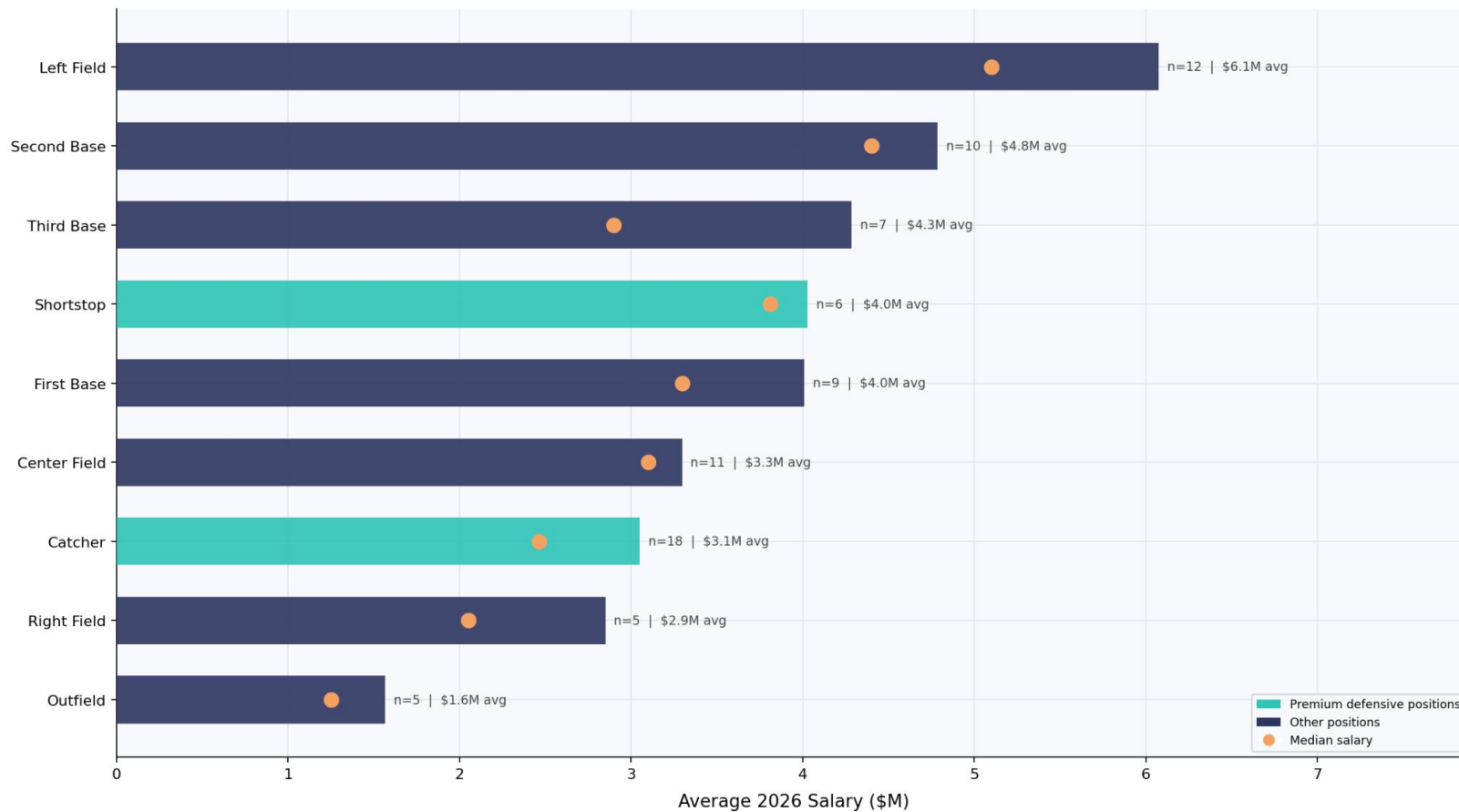
Distribution of Games Played in the Season for Individual Players



Distribution of Career Seasons Played for Individual Players

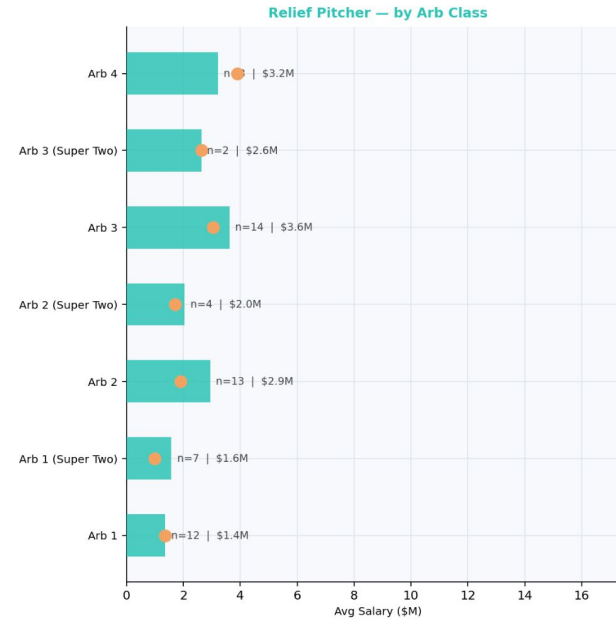
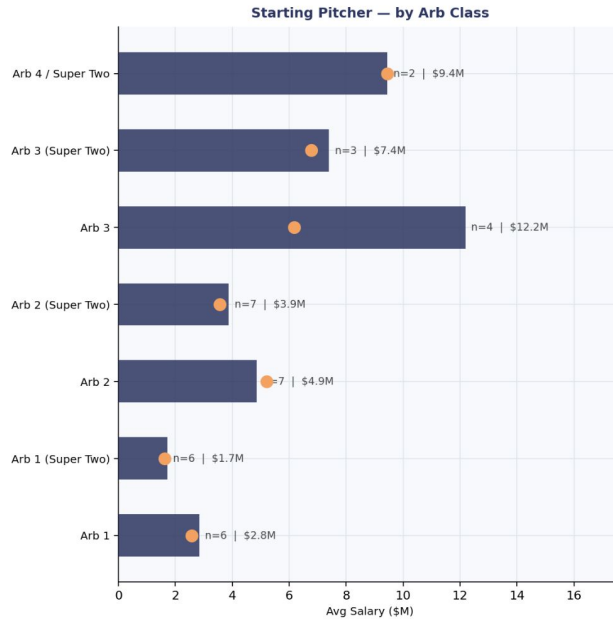
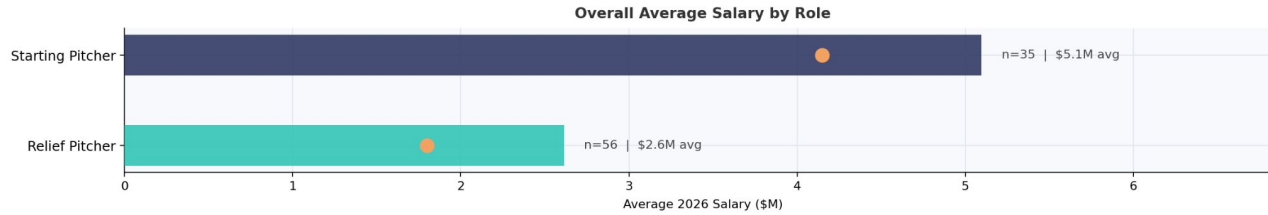


2026 Arbitration Salary by Position — Hitters



Three catchers in the top 5 most overpaid — panels reward position scarcity above recent performance.

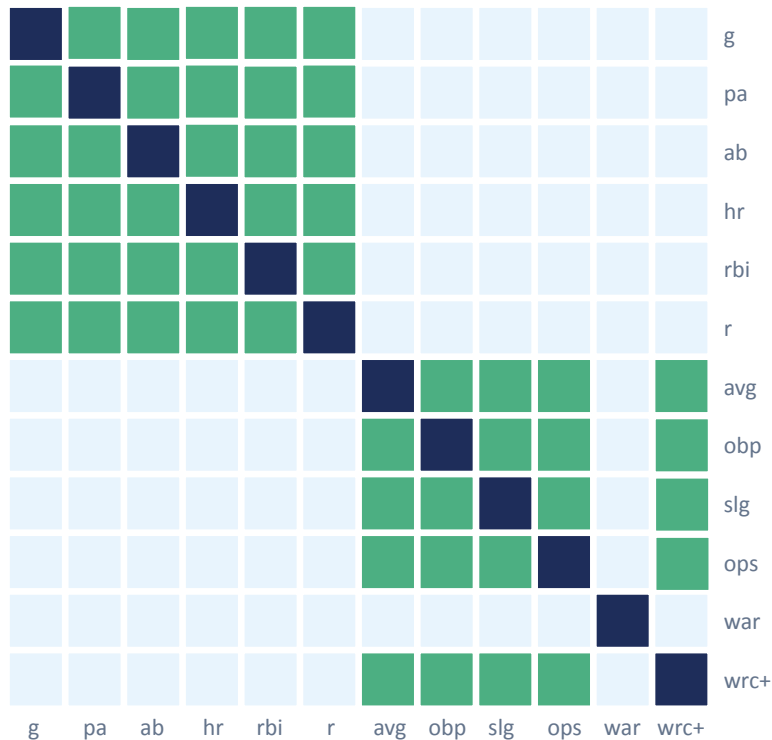
2026 Arbitration Salary by Role — Pitchers



Only arb classes with ≥ 2 players shown in breakdown panels. Starting pitchers earn significantly more across all arb years.



Highly Correlated Data



- Many variables highly correlated (>0.85)
 - G/PA/AB/HR/RBI/R
 - AVG/OBP/SLG/OPS/WRC+
- Some of these statistics measure virtually the same thing
- Feature selection measures taken (as shown later)

Leave-One-Year-Out Cross-Validation

Shuffling years, as would be the case in standard k-fold cross-validation, would let the model train on 2025 data to predict 2023. This leaks future salary information

Fold A	Train on: 2024 + 2025	Test on: 2023
Fold B	Train on: 2023 + 2025	Test on: 2024
Fold C	Train on: 2023 + 2024	Test on: 2025

Model trains on all 3 years → Predictions applied to 2026

Baseline Model Results

Hitters

Model	RMSE	MAE	MAPE	R ²
Linear Regression	\$1.663M	\$1.193M	44.3%	0.840
Ridge ($\alpha=1$)	\$1.501M	\$1.054M	36.1%	0.871
Lasso ($\alpha=10k$) ★	\$1.467M	\$1.039M	35.1%	0.878

Pitchers

Model	RMSE	MAE	MAPE	R ²
Linear Regression	\$1.620M	\$0.939M	45.8%	0.723
Ridge ($\alpha=1$)	\$1.665M	\$0.957M	48.4%	0.709
Lasso ($\alpha=10k$) ★	\$1.620M	\$0.917M	44.5%	0.724

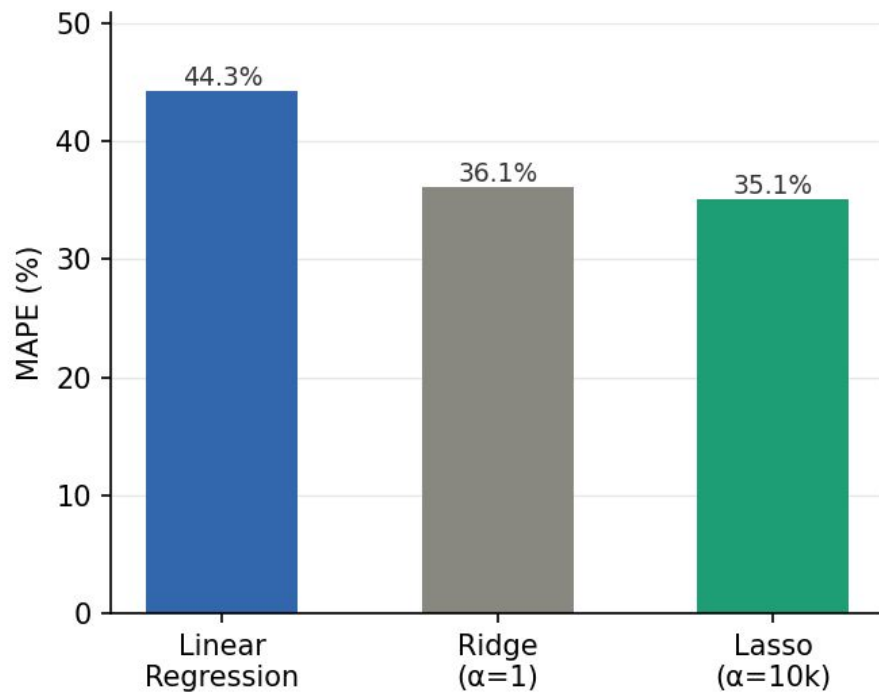
MAPE is the best metric to analyze, as it scales with salary, unlike RMSE or MAE

By MAPE (and R²), zeroing coefficients with Lasso outperforms shrinking coefficients with Ridge or pure OLS

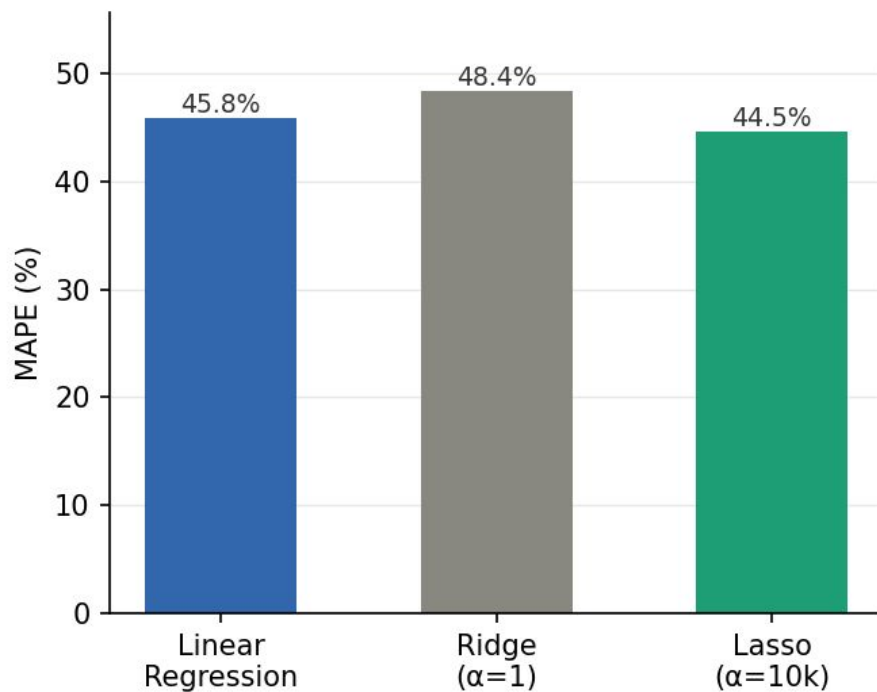
Baseline models: average MAPE across LOYO-CV folds

Mean absolute percentage error. Lower is better.

Hitters



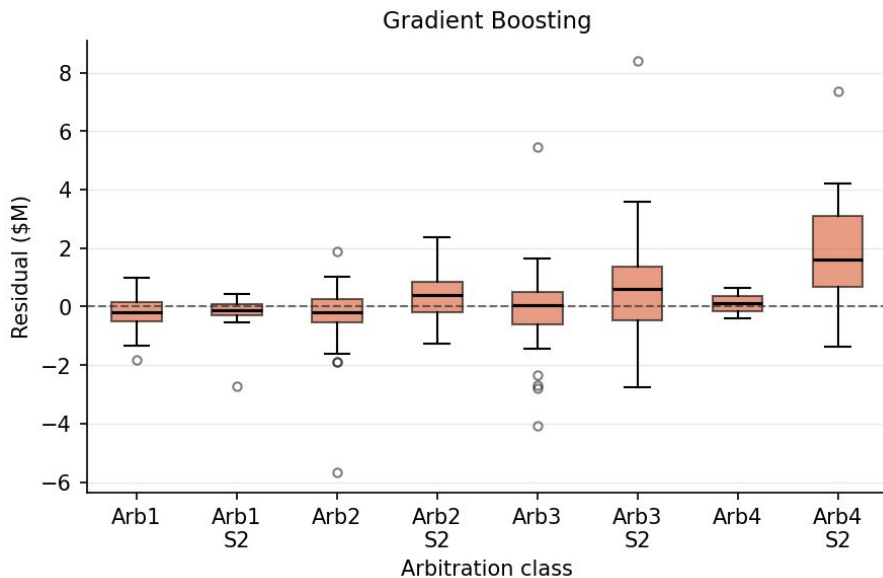
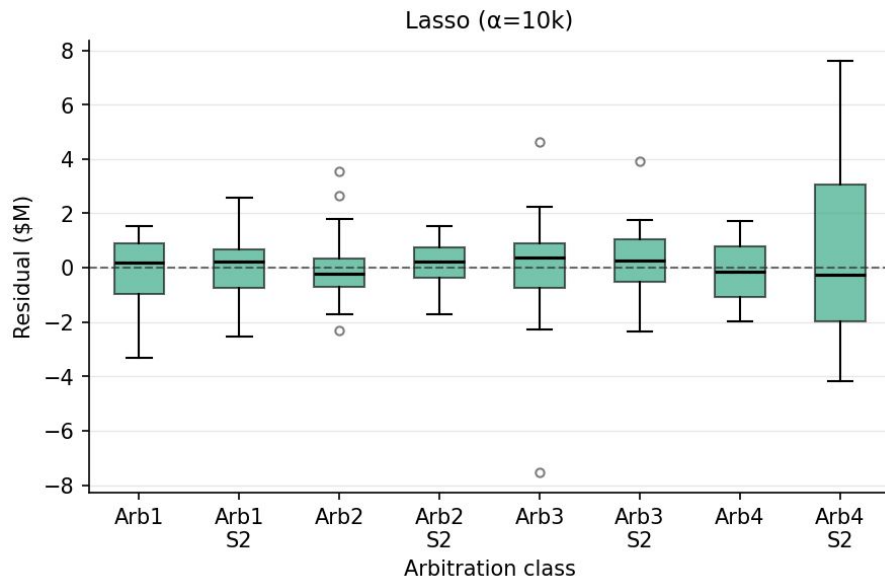
Pitchers



Model Accuracy by Arbitration Class

Residuals by Arbitration Class — Hitters

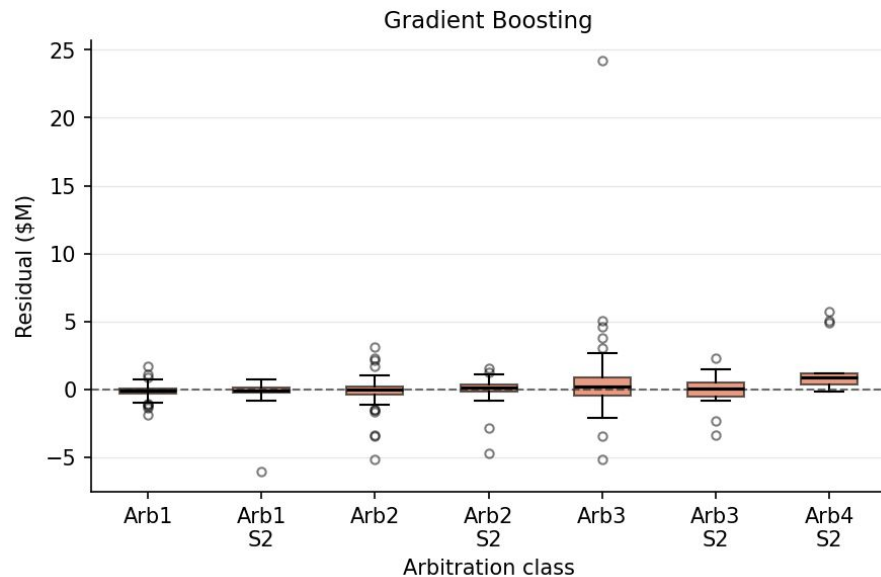
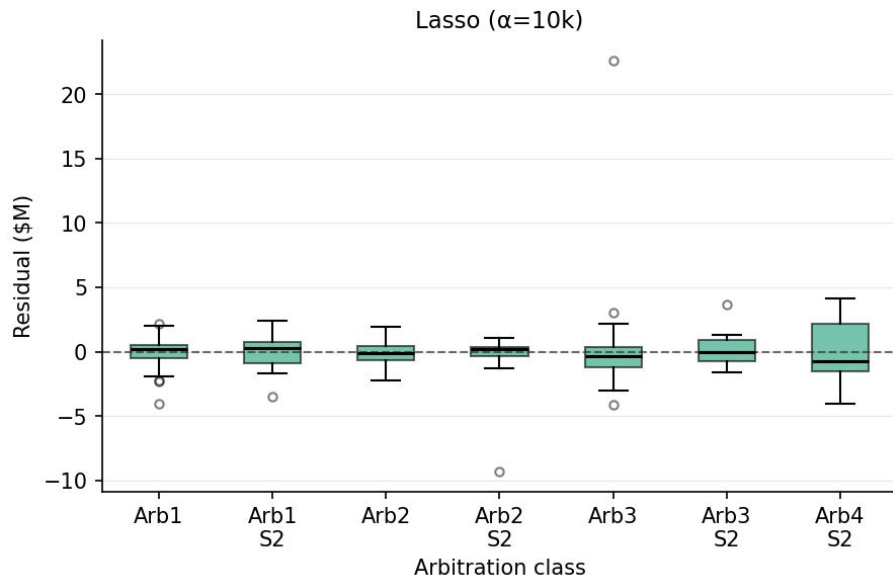
Residual = Actual – Predicted. Positive = model underestimated salary.



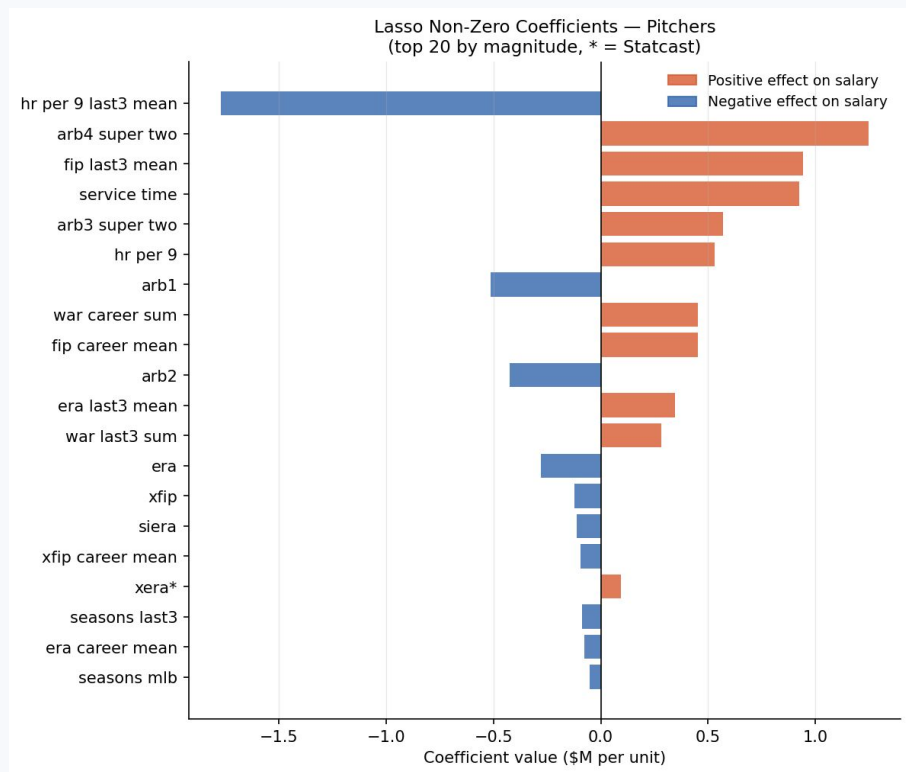
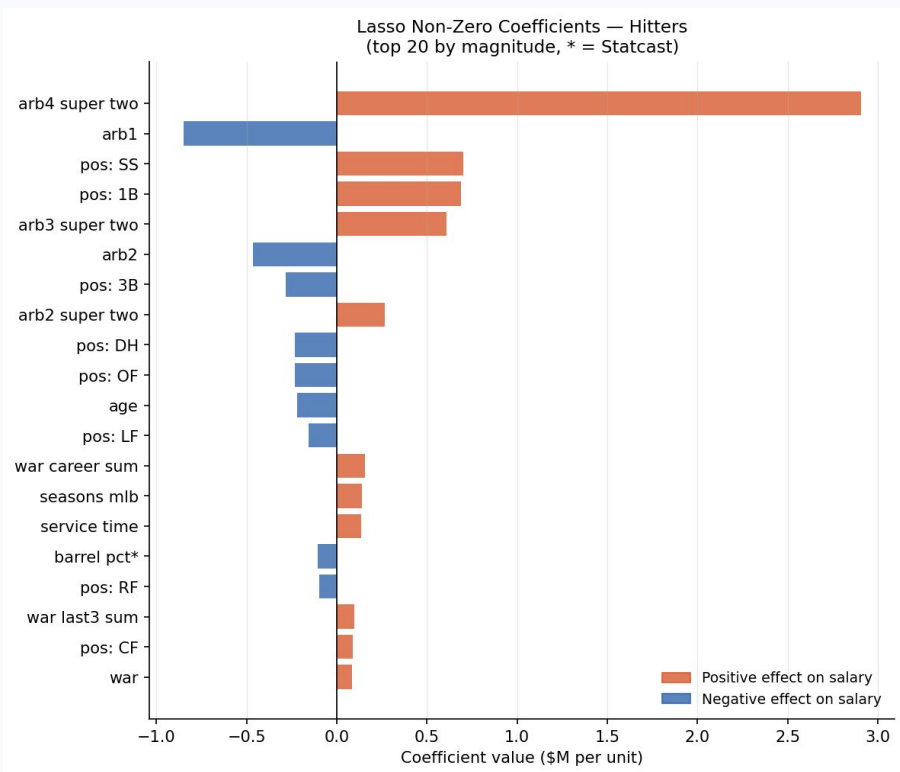
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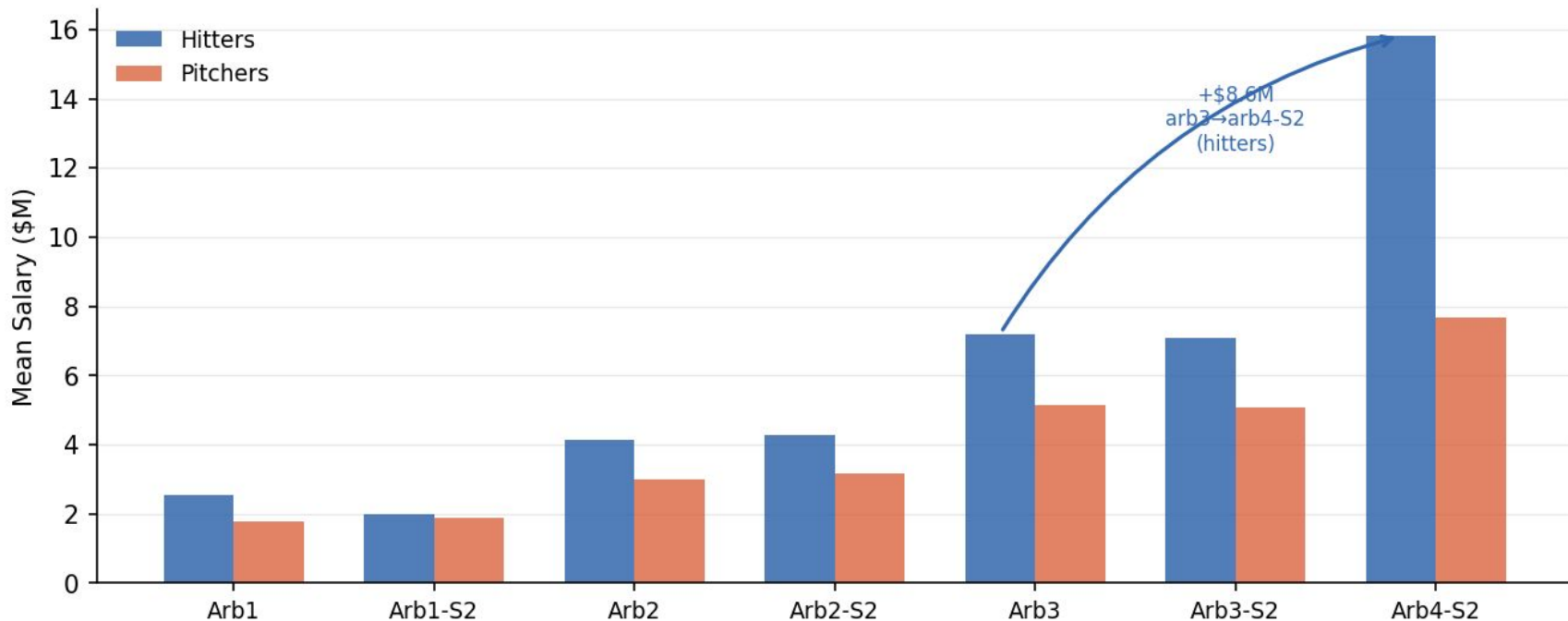


Variables Driving Arbitration Salary



Impact of Service Time (and Arb Class)

Mean Arbitration Salary by Eligibility Class (2023–2025 Training Data)
Salary escalates sharply with service time. Super-two players (S2) enter the arb pool one year early.



Prediction Characteristics



2023-2025 as training years, blind to 2026



85 arbitration-eligible hitters predicted



92 arbitration-eligible pitchers predicted



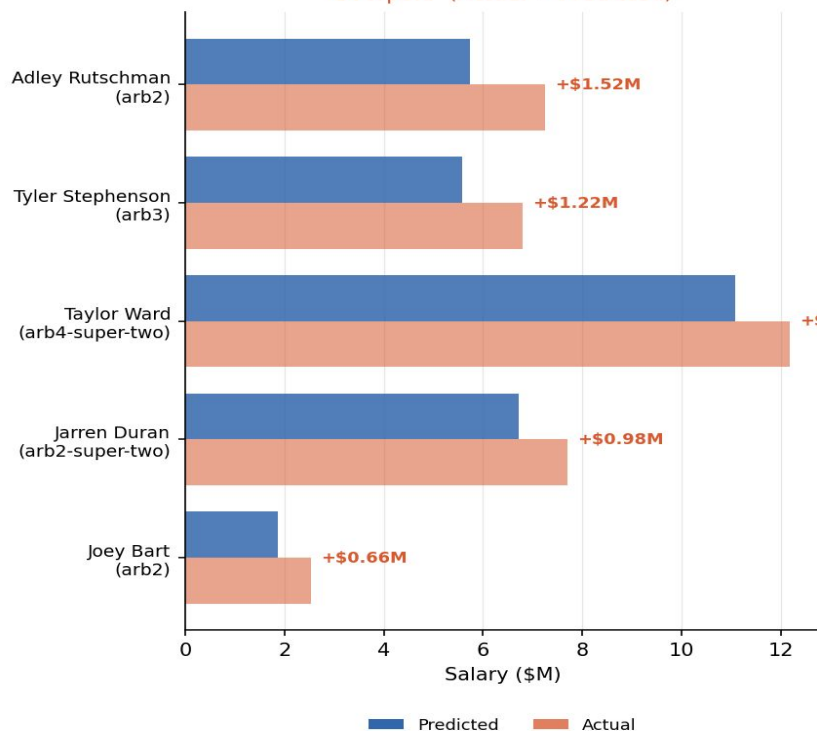
Estimates what should be awarded, likely closer to “midpoint”, not what the player or team file for

Results: 2026 Overpaid and Underpaid Hitters

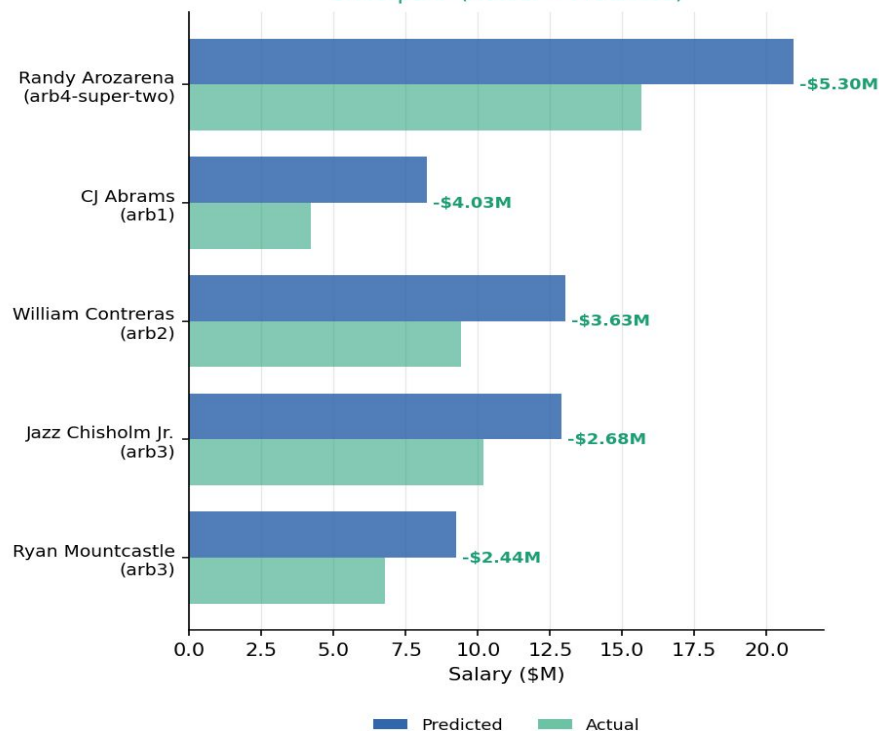
2026 Hitters: Most Overpaid and Underpaid vs. Model Prediction

Each player shows two bars: Predicted (top) and Actual (bottom).

Overpaid (Actual > Predicted)



Underpaid (Actual < Predicted)

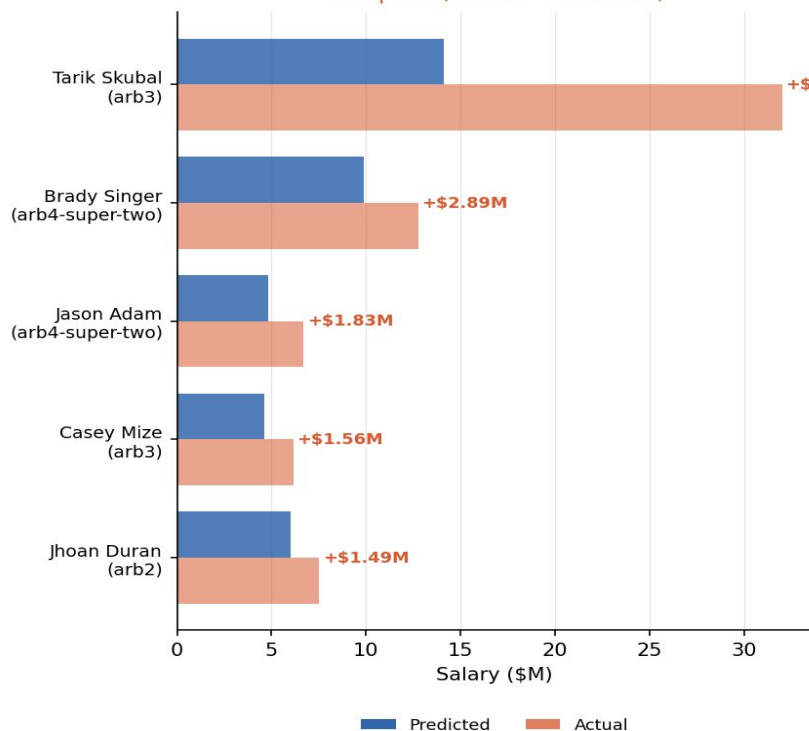


Results: 2026 Overpaid and Underpaid Pitchers

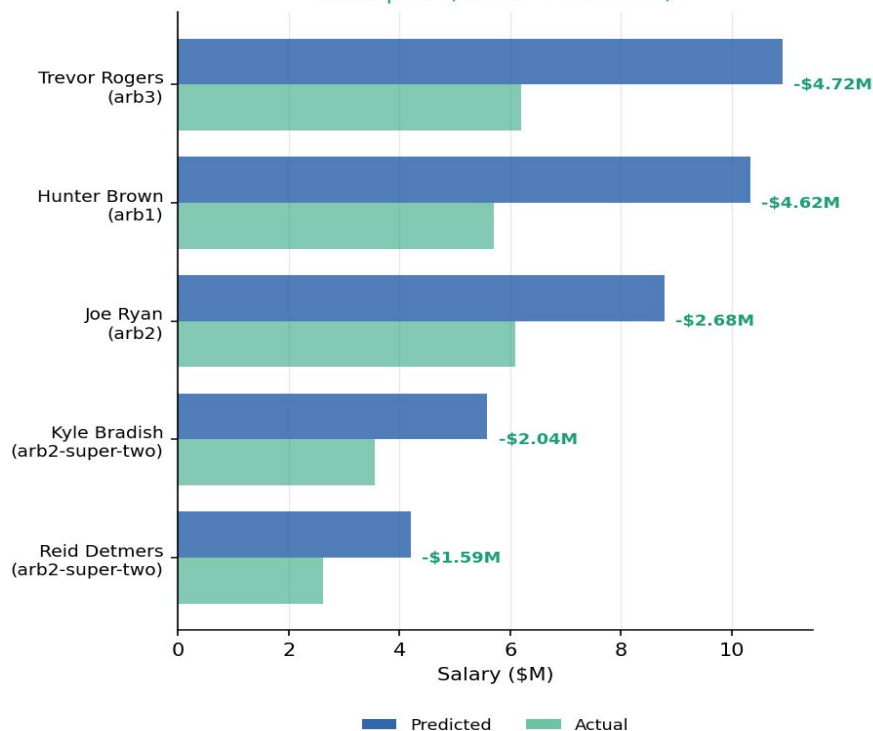
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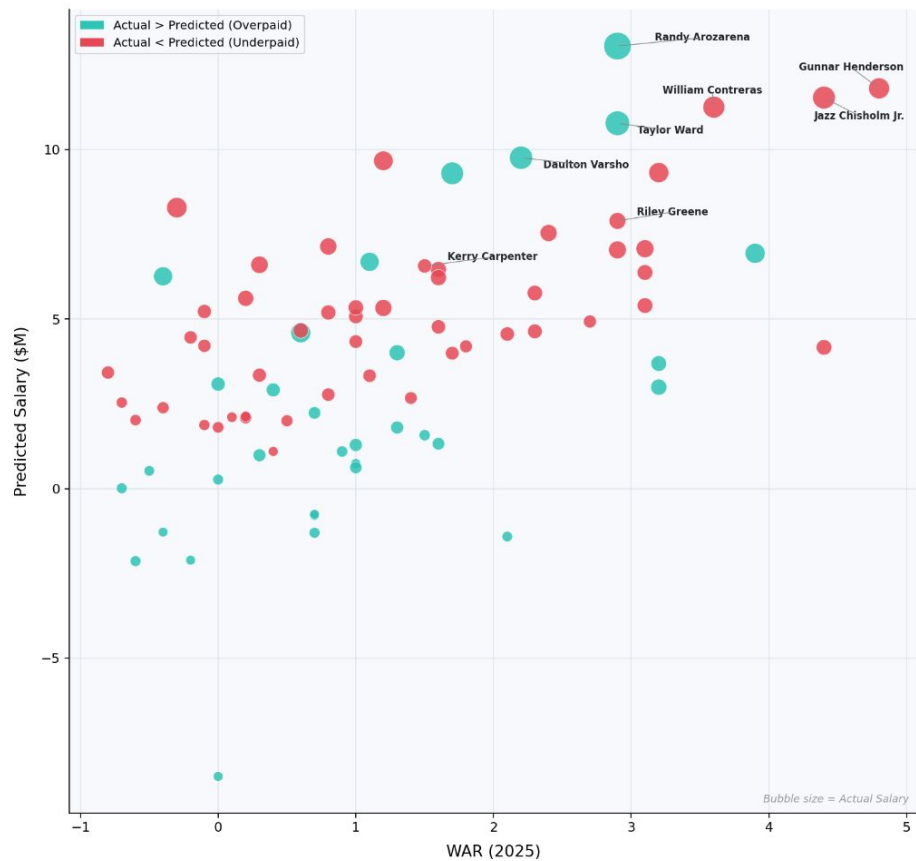
Overpaid (Actual > Predicted)



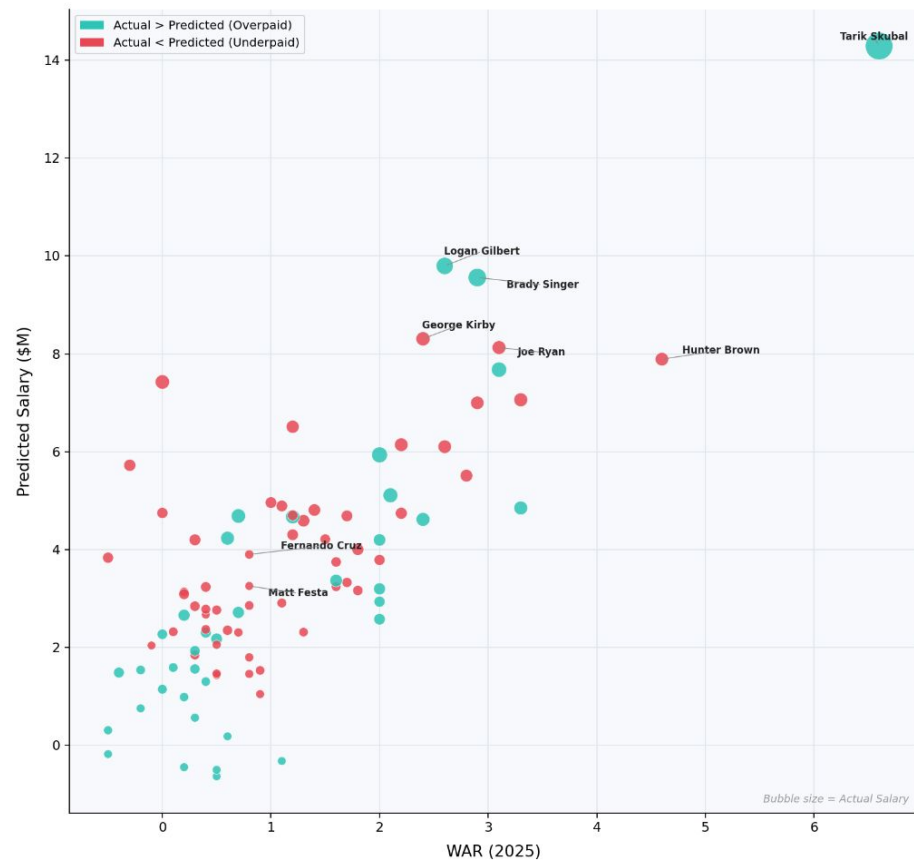
Underpaid (Actual < Predicted)



Hitters — WAR vs. Predicted Salary



Pitchers — WAR vs. Predicted Salary



Offensive Production vs. Salary — Hitters 2026

Bubble size = Model Predicted Salary



Pitching Performance vs. Salary — 2026 Arbitration

Bubble size = Innings Pitched



Model Limitations

Error Impacts Player Predictions Differently

An Arb1 player sees a greater effect from \$1M error than an Arb2 or Arb3 player being paid more

More Variables Exist Than Reflected

While the model takes a ton into account, baseball has many nuances that are not exhibited

Further Historical Information Available

Three years is the industry standard for projections, but more examples may improve the model

Real-World Model Value

Team Front Offices

Can identify extension opportunities to avoid arbitration, or target underpaid players in upcoming FA

Player Agencies

Can challenge a lowball filing with a strong objective argument, or more easily identify player comps

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<https://www.mlbtraderumors.com/2011/10/mlb-trade-rumors-arbitration-projections.html>
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